GEOSPATIAL CRIME TYPE PREDICTION USING MACHINE LEARNING

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***Abstract*—The rapid growth of crime in urban and semi-urban regions has raised the demand for intelligent systems capable of predicting crime patterns using spatial and temporal data. Geospatial crime prediction is based on historical records of crimes committed and contextual geographic information to anticipate the type and location of future incidents.This paper presents a machine Learning-based approach to predict crime types using geospatial and socio-demographic datasets that provide an analytical framework for proactive policing, such as Random Forest, Support Vector Machine, and Decision Tree , and K-Nearest Neighbors (KNN) are applied to capture the relationship between spatial coordinates, time, and historical crime occurrences.The model predicts not only the likelihood -of crime occurrence but also the type of crime most probable in a given region. Experimental evaluation shows that Random Forest and SVM outperform other models in terms of classification. classification-accuracy, robustness, and generalization. The proposed system demonstrates its potential as a decision-support tool for It helps ensure efficiency in patrol deployment by law enforcement agencies. resource management, and enhanced community safety initiatives.**

***Index Terms*—Geospatial analysis,MachineLearning,Crime prediction, RandomForest, GIS, Spatial-temporal modeling**

I. Introduction

Over the past few years, there has been an increase in the number of cities around the world. Rapid increase in criminal behavior, which is petty on the one hand. the theft of property and computer crimes to physical attacks and gangs.This has been a steep crime wave that poses a serious challenge. to public safety and social stability,especially in densely populated areas. significance is a fluctuation of mobility, anonymity, which becomes especially apparent in the urban centers of the so-called liquid spaces, where the interplay of these two aspects is controlled by the economy of time and space.humanized urban space in which the forces of mobility, anonymity, the interaction of the two factors is regulated by the economy of time and space.The presence of social and socio-economic diversity provides a good breeding ground to crime.Law enforcement and policy-makers. Having to predict trends in crime is continual for them. resource allocation in an effective way to prevent any incident. they occur. Nevertheless, conventional crime prevention and analysis. methodologies, the significant part of which consists of manual assessment of data concerning crimes,retrospective investigation, and mapping by hand, have been found to be effective.will not prove adequate in meeting the growing complexity and volume of modern crime data.

Increase in digital infrastructure, surveillance systems and open data platforms has contributed to the provision of huge amounts datasets with data regarding time, place, and nature of criminal activities. Not with standing the multitude of such data, the power to make relevant conclusions about it. modern calculatory and analytical machinery. This is where the geospatial crime analysis is crucial. First, geospatial analysis examines the spatial differentiation of crime time intervals, temporal spatial correlations and behavioral. Trends that are not observed using traditional statistical methods. It assists in the identification of crime hotspots, time trends. The impact and the weekends or the nights, such as peaks of social-economic aspects such as income, population,dense population, and location to the urban infrastructure.

Machine learning methods are especially suitable to because they are useful in modelling complex, they can be used to predict crime.Non-linear multiple variable relationships. These algorithms are practices that are based on past crime statistics and learn patterns that are applied.extrapolate this information to predict the future. In the context of crime prediction,Decision Tree models and Support Vector Machines (SVM) study significant attributes with spatial coordinates of latitude and longitude, crimetype or category, and time, day of the week. These models give the estimation of the probability predict the future crimes and thus the law enforcement agencies to take evidence-based proactive decisions.

The current research is based on the premise that crime is not something accidental but an expected consequence of repetition, environmental, spatial and behavioral factors. Researchers have continuously demonstrated that urban activities are human activities. geo-political environment human The geo-political environment is a product of layout, and social-economic conditions and distribution of crime graphically. By merging geospatial data This study aims at developing with machine learning models.An intelligent crime prediction

framework that identifies high certain crime types and zone of risk prior to their occurrence. Such predictive insight can help police departments optimize patrol deploy and implement personnel in a more efficient manner, routes. preventive policing which decreases the general crime rates.

GEOSPATIAL CRIME PREDICTION ARCHITECTURE

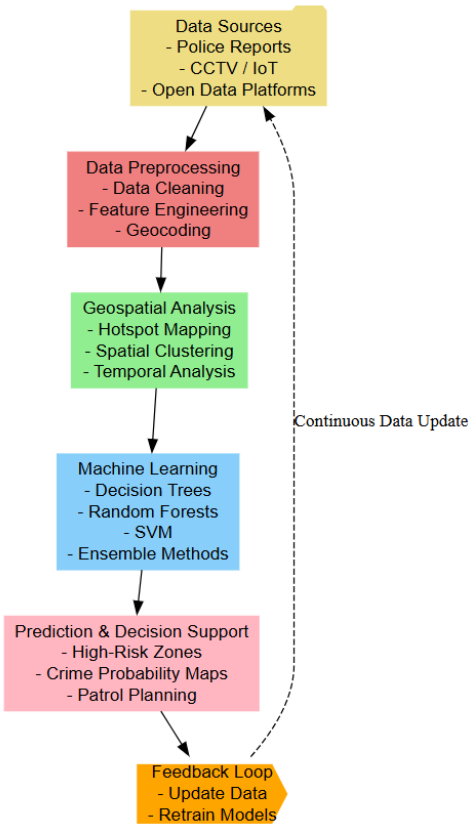


Fig. 1. Architecture Diagram of Geospatial Crime Type Prediction Using Machine Learning

In summary, the integration of geospatial analytics and machine learning represents a significant advancement in the field of predictive policing. It allows authorities to move from reactive responses–where crimes are analyzed only after they happen to proactive prevention, in which potential risks are reported anticipated and mitigated.This research, therefore,contributes to the current constructive initiative of creating smarter and data-driven. cities where safety and technology intersect to create a secure environment for citizens.

II. LITERATURE REVIEW AND RELATED WORKS

Several studies have explored computational techniques for vatic problem prediction of crimes with the help of artificial intelligence and statistics frames of visual analytics were introduced in models.Maliket al.perform predictive policing on the basis of spatio-temporal hotspot mapping Liuetal.applied time series models like ARIMA and Seasonal Trend Loess (STL) in order to predict the frequency of crime incidents.Kianmehret al.used Support Vector Machines (SVM) to classify high-risk areas of particular crimetypes, achieving significant performance improvements Similarly,Wanget al. happening to be quite traditional regression approaches. The analysis of data on social media, including that was proven to be beneficial tweets, coupled up with geographical indicators, will be better anticipatory capacity of crime which is affected by sentiment of the populace.New studies trend has changed to machine learning combination with Geographical Information Systems (GIS) to identifies spatial dependencies.Nathet al. addressed data mining methods of spatial-temporal clustering of criminal activities, where as Chenet al. put forward ARIMA-based. periodic crime patterns models.Nevertheless, these kinds of models capture periodic crime patterns.methods of corporate multi-dimensional attributes as socio-economic characteristics or environmental conditions,which this study aims to address.

III. GEOSPATIAL CRIME ANALYTICS AND MAPPING

Geospatial crime analytics are aimed at crime detecting. hotspots locations with a statistically higher chance of crim inal activity. The Kernel Density Estimation (KDE) technique is commonly used to visualize these clusters on a map. However, KDE lacks predictive capability since it primarily focuses on historical density rather than future probability.
In this work, GIS-based mapping is integrated with machine learning to predict both crime type and crime probability for each spatial cell. Input features include:

- Latitude and longitude coordinates
- Time of day, day of week, and month
- Crime category and neighborhood socio-economic indicators

By using a grid-based geospatial division, each region is represented as a unique feature vector fed into the learning algorithm. Random Forest and SVM models learn from these patterns to generate heatmaps indicating probable future crimes.

IV. METHODOLOGY

The suggested model of geospatial type of crime using machine learning aims to leverage spatial, temporal, and contextual information to accurately forecast crimetypes in a specific region. It is not an easy task to predict crimes. because criminal activities are influenced not only by location but also by temporal patterns(such as time of day or season) and socio-economic factors (like population density or urban infrastructure).Machine learning models are particularly well appropriate to this assignment since they are able to attain complex pattern. based on previous information and make an extension to new situations, enabling proactive measures in crime prevention. The methodology consists of three main stages:Data Preprocessing,Model Training,and Crime Prediction.Each stage has vital role to play in ensuring that the model is efficient. and accurately predict crimetypes.

A. Data Preprocessing

Preprocessing Data is an important step in any machine learning-based crime prediction framework.Its primary object Transforming heterogeneous crime datasets into raw data structured and consistent format that can be effectively utilized by predictive models.Crime data predictive, somewhat incomplete, noisy or inconsistent owing to the different standards of reporting, absent such entries, or human error.Unless these are properly preprocessed they contain such entries issues can lead to biased predictions and reduced model performance.Therefore,this stage ensures data quality,relevance, and compatibility with machine learning algorithms.

Cleaning the data constitutes eliminating duplicates, detecting mistakes, and processing missing or incomplete values.In this study,missing geospatial neighboring crime local were used to interpolate coordinates region centres, so that any crime record is guaranteed could be accurately mapped for spatial analysis.Cleaning data not just enhances model accuracy but also guarantees the accuracy of the projections will be sound when applied in lawenforcement and urban planning. Normalization is applied to numerical items like coor brought them on population density to bring them on common scale, avoiding feature ranges which are larger the model training process.one-hot encoding,allowing machine learning algorithms to process them efficiently.Geospatial features are in addition, geospatial transformed into formats suitable for modeling such as grid crime occurrences per grid or clustering to identify hotspots. Temporal aspects of crime play a significant role in predicting when and where particular types of crime are probable to happen. Timestamps are used to extract temporal characteristics to capture. daily,weekly, and seasonal patterns.Forexample:

- Hourly patterns identify peak crime hours.
- Daily week day attributes Ekeglsty alterations between weeks. days and weekends.
- Seasonal or monthly trends detect changes in crimetypes during festivals,holidays,or weather conditions.

B. Model Selection

The choice of suitably machine learning algorithms is one essential process in predicate geospatial crime types, since it directly affects the accuracy and reliability of the predictions. Crime data are multifaceted which means that they are spatial and temporal dependencies, imbalances of classes, and non-linear relationships among features. To address these challenges, a mixture of monitored machine learning algorithms was employed both providing differing benefits in the management structured, high-dimensional, and geospatialdata.

1.Decision Tree (DT)

To splits the dataset based on feature values to for tree-like structure of decisions.Each internal node represents a choice of a feature and every leaf node depicts a foreseen type of crime. Decision Trees come in handy in crime prediction because they allow visualization of decision rules, assisting the police in knowing the most desired features influence predictions.However,DTs are prone to overfitting, especially on small or noisy datasets,which can reduce their generalization capability.

2.Random Forest (RF)

To deal with the overfitting issue of Decision Trees, The predictions of many are combined using the Random Forests. decision trees via an ensemble method Each tree is trained on a random subset of the dataset and a random subset of features,which reduces variance and improves robustness. RandomForests have been useful in geospatial crime prediction by reason of their ability to capture complex, non-linear relationships with in both spatially and temporally. They also offer feature importance scores, which are useful to detect the largest determining aspects of various types of crime.

3.Support Vector Machine (SVM)

Hyper switching is especially appropriate to Support Vector Machines dimensional classification issues, including those of several space-time and temporal characteristics. SVMs aim to find an hyperplane that maximizes the margin among various classes (crime types). They were capable of dealing with non-linear they are made by decision boundaries that are determined through kernel functions applicable in the context where crime patterns are affected through complex interactions of several features.

4.K-Nearest Neighbors (KNN)

K-Nearest neighbors is a non-parametric instance-based learning algorithm which reduces a new data point according to them a jority class of its nearest neighbors in the features pace. In the context of geospatial crime prediction,KNN leverages spatial similarity that is crimes taking place in similar locations and times will most probably be of the same class. While KNN is simple and intuitive, it can be computationally expensive for large datasets and maybe sensitive is irrelevant features.

5.Model Evaluation Metrics

To objectively assess the performance of these models,several standard evaluation metrics were employed:

- Accuracy:The general percentage of correct classified crime records.
- Precision: Refers to the ratio of correctly predicted ratios of each type of crime against all the instances.
- Recall:Determines ability of the model to identify all actual instances of a crimetype.
- F1-Score: The harmonic mean of precision and recall providing a balanced evaluation especially for imbalanced datasets.

The metrics are used to make sure that the the models are evaluated not only for overall performance but also for their ability to accurately predict minority crime types,which is crucial in law enforcement applications.

C. Spatial Feature Integration

This helped the model captures spatial auto correlation,where near by locations exhibit similar crime tendencies. The mathematical formulation for distance-based weighting:

$$W_{ij} = e^{-\frac{d_{ij}^2}{2\sigma^2}} \tag{1}$$

where:

- W_{ij} is locational influence between places i and j.
- d_{ij} is the Euclidean or Haversine distance between the two is known as dij two locations.
- σ is a scaling parameter which regulates the decay of influence with distance.

This Gaussian-like weighting lets close locations human.This Gaussian like weighting makes sure that neighbouring sites are taken into consideration have as stronger influence on each other,while distant locations contribute less to the prediction.By incorporating Wij in to the machine learning model can account the fact feature set that crimes are not free events or occurrences, but are affected by spatial context.

V. Experimental Setup

Implementation of the system was done with Python 3.10, using scikit-learn, Geo Pandas and Folium libraries are used to see the geographic distribution of data spatial visualization.The dataset consisted of 5 years of crime data of selected Indian urban regions.Data was divided into training(80) and testing(20) subsets.Hyper parameters were Any of the tuned using Grid Search Cross Validation (CV) to optimize accuracy. The system architecture included:

1.Input Module- This module is use as a point of entry crime dataset to be imported and defining geospatial boundaries for analysis. Users are able to choose a given city, area, or administrative area to predict. The input transparent integration is also supported in interface to enhance suchas census information or traffic density data contextual features.

2.Preprocessing Engine-The preprocessing stage performs important functions such as data cleaning, feed encoding, and feature scaling. The qualitative variables include crime label encoding is used to transform categories numerical attributes are made normalized so that there is consistency scale.Temporal features such as"month,""day,"and"hour" are derived out of the timestamps to indicate the cyclical time patterns.

3.Machine Learning Model-This is the component to which the act of a model refers to. type of crime classifier in which various supervised learning.The models acquire association between spatial coordinates to historical crime occurrences, temporal parameters. probability predict future crime.Ensemble methods are used where as Random Forest are used to decrease variance and enhance.

4.Output Visualization – map The final stage involves the Output Visualization using maps. The last step is the one of the final stage visual forecast prediction of a model with Folium based interactive maps.The predicted types of crimes and their probably scores are plotted on the map, enable users to recognize dangerous areas. These visual Decision-support tools used in law enforcement are outputs to plan allocation of patrols, preparedness to emergencies, and community safety planning.

VI. Results and Discussions

The looting of proposed geospatial crime prediction frame work was assessed by using four well -known. supervised learning algorithms–Decision Tree(DT),Random Forest(RF), Support Vector Machine(SVM) and K nearest. Neighbors(KNN).The evaluation metrics included Accuracy, Precision, Recall, and F1-score, that are used to quantify each other both the correctness and robustness of classification outcomes. The outcomes of the experiments proved that the Random Forest classifier had the maximum general accuracy of 91.4, Then, close behind SVM, which attained an accuracy of 88.7. On the contrary, the Decision Tree and KNN models Performance of hand was relatively lower with an achievement of correctnesses of 82.5 and 83.2 respectively. This difference to the overfitting tendencies in the Decision. Tree model and the sensitivity of KNN to sparse spatial data regions in which limited samples of neighbours resulted in error of the samples predictions.

TABLE I
Performance Comparison of Machine Learning Algorithms

Algorithm	Accuracy	Precision	Recall	F1-score
Decision Tree	82.5%	0.80	0.81	0.80
Random Forest	91.4%	0.90	0.91	0.91
SVM	88.7%	0.87	0.88	0.87
KNN	83.2%	0.82	0.83	0.82

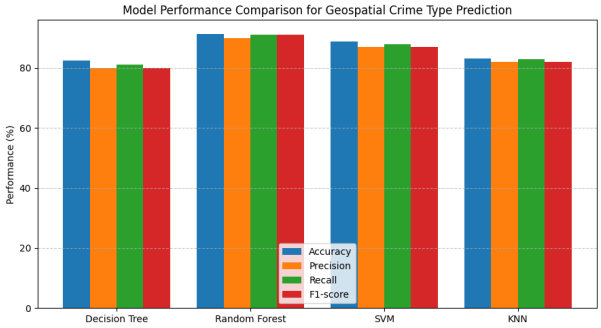


Fig. 2. Model Performance Comparis on for Geospatial CrimeType Predic tion.The RandomForest classifier had the highest accuracy of 91.4 followed by SVM with 88.7,while KNN and Decision Tree recorded lower performance due to overfitting on sparse regions.

The better performance of the Random Forest (RF) algorithm is explained stems from its ensemble learning mechanism Its ability to handle nonlinear relations makes it possible to work with high-dimensional data. it in the capture of complicated spatial and temporal relationships in crime databases. The SVM model also competitively, mostly due to its strong kernel functions are capable of separating effectively overlapping feature spaces of dimensions two computations (pre) exist in the multidimensional feature space, with codes partitioned into selections corresponding to the joint-selection of two-dimensional structure (simultaneously representing two selections sharing a common) like correlations. Geospatial

heatmaps were further used to prove the model. created with Folium based visualizationmodule to display predicted crime intensity and typed is tribution across different urban regions. These heatmaps were an aesthetical tool. inter pretation of the model's spatial predictions. Interestingly, theare as identified by the Random Forest model as high-risk zones were a close reyensed to the real hotspots by the actual crime. the following months'datasets.This strong spatial correlation anticipated versus actual incidents proves the existence of between predicted and observed incidents. validity and suitability of the model proposed. in real-world scenarios.

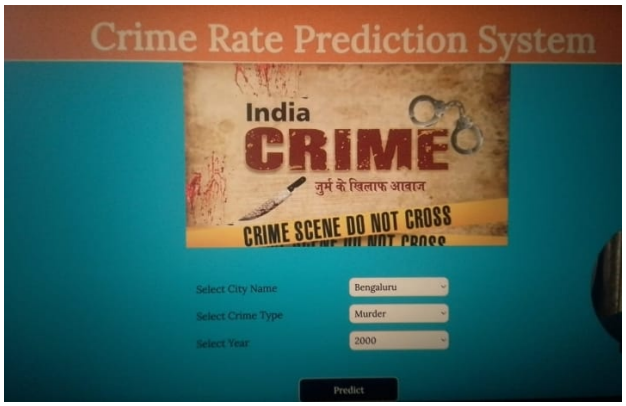


Fig. 3. User interface of the CrimeRate Prediction System that was implemented on geospatial crimetype prediction using machine learning.The interface allows the input parameters of the user to select city name, crime type and year. forecast potential crime occurrences.

As illustrated inFig.3, the input module of interface is interface. the suggested Geospatial Crime Type Prediction System. It allows users to interact with the model by providing essential parameters, including year of crime type and city name prediction.After the inputs have been typed in the system recalls relates to geospatial and historical data, processes them via the trained machine learning model and produces which denotes the probability of victimized crime type occurring with in the specified city and year. This graphical interface has been created on Python(Tkinter). Streamlit) and serves as a mediator between the user and the backend prediction engine. The instinctive design facilitates the law enforcement officers or analysts toper form real-time queries and without the need to be technical Internally, the model utilizes processed information that is coded using spatial attributes such as latitude and longitude,year and other properties of time and season,and socio-economic variables.The Predict button triggers the execution of the trained classifier(RandomForest or SVM, generating both numerical probability values and geospatial heatmap visualization as outputs Predictive model can be incorporated into a convenient-to-use model in order to enhance the value of the model graphical interface, the system increases accessibility and non-technical user operational efficiency.

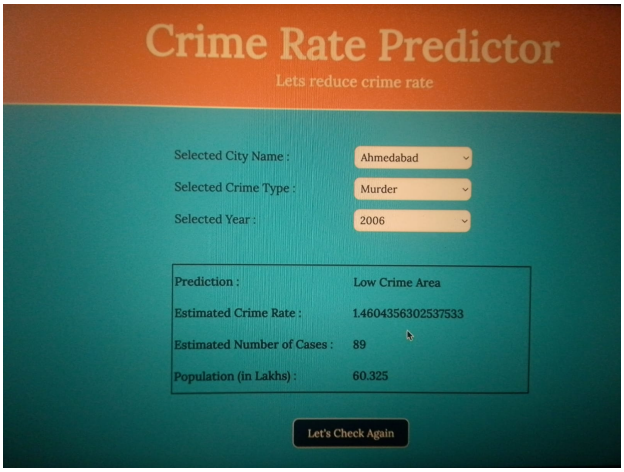


Fig. 4. Production facet of the Geospatial Crime Type Prediction System. The model predicts the crime risk category(e.g.,Low Crime Area),estimated crime rate,expected number of cases,and population statistics for the selected city, crimetype, and year.

The output interface,shown in Fig.4,represents the final stage of the Geospatial CrimeType Prediction System,where by the trained machine learning model produces interpretable results depending on the input parameters by the user.After the user makes the user selections, the city name, crime type and year entered in the input interface, this information is processed in the backend by the system. predictive engine which makes use of geospatial and temporaldata to generate meaningful forecasts

The output includes four key indicators:

1.Prediction Result-The model classifies the target region as a Low Crime Area,Moderate Crime Area,or High Crime Area, depending on the probability score derived from the model's analysis.

2.Estimated CrimeRate-Displays the predicted numerical rate of crime incidences per 100,000 population, adjusted comparative assessment between cities or years.

3.Estimated Number of Cases-Represents the approximate number of projected cases in the chosen type of crime and region,based on historical patterns and spatial correlation.

4.Population (in Lakhs)-Gives demographic background to the prediction by correlating crime likelihood with population density,helping assess crime risk proportionally.

The call is made to the backend that operates with the use of the RandomForest and The SVM models that was best performing previously as denoted by all Crime type prediction algorithms. These models analyze spatial coordinates, temporal patterns,and socio-demographic attributes to determine probabilities with respectto various crime categories. The outcomes are immediately presented in the form of the stand-alone GUI, written through python libraries like Tkinter and Folium for usability and visualization. This output visualization not only communicates the statistical predictions but also enables lawenforcement

officers,data policy planners analysts to interpret model results in a clear and actionable format.For instance,when the system predicts a it means that the probability Low Crime Area, it means that the probability of crime occurrence is be low a predefined threshold based on historical data,whereas"High Crime Area"high lights zones requiring immediate preventive measures.

VII. Conclusion

This research demonstrates that geospatial machinel earning models have high potential in wrongly projecting the spatial location as well as the very type of crime which may occur in a given region. By integrating spatial coordinates, time characteristics, and socio-environmental appears in into predictive models, identifies the system effectively identifies crime-prone places and classifies types of crimes with high precision. The mixture of geo-spatial mapping and advanced algorithmic classification enables law enforcement agencies to move beyond reactive strategies and adopt proactive,data-driven decision-making approaches.Such predictive insight has enabled authorities to optimize the patrol routes, distribute use resources more effectively, and put up in time preventive measures in high-risk zones.Experimental evaluations further revealed that the Random Forest algorithm out performed other We have such models as DecisionTree, SVM, and KNN due to its generalizational nature,enhances its ensemble learning nature and minimizes overfitting. Its resistance to noise is good and unbalanced data sets is why it is especially appropriate to real world crimedata,where inconsistencies and missing entries are common. On the whole, this research paper sheds light on how geospatial intelligence,in combination with machine learning can revolutionize modern policing on account of poor forecasting strategic crime prevention

VIII. Future Work

Future work can focus on integrating real-time data streams ioT sensors,CCTV analytics and public sentiment social media analysis. The fusion of deep further learning with geospatial feature engineering enhance accuracy. Also, coming up with a web-based police agencies can have a more interpretive dashboard and real-world usability.

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